

In the following formulas parameters n , m are integers, and $0 < a \in \mathbb{R}$ and $b \in \mathbb{R}_0$:

Trigonometric identities

Here α and β are real numbers.

$$\sin 30^\circ = \frac{1}{2}, \quad \sin 45^\circ = \frac{\sqrt{2}}{2}, \quad \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\sin^2(\alpha) + \cos^2(\alpha) = 1$$

$$\sin(\alpha \pm \beta) = \sin(\alpha) \cos(\beta) \pm \cos(\alpha) \sin(\beta)$$

$$\cos(\alpha \pm \beta) = \cos(\alpha) \cos(\beta) \mp \sin(\alpha) \sin(\beta)$$

$$\sin(2\alpha) = 2 \sin(\alpha) \cos(\alpha)$$

$$\cos(2\alpha) = \cos^2(\alpha) - \sin^2(\alpha)$$

$$\tan(\alpha) = \frac{\sin(\alpha)}{\cos(\alpha)}, \quad \cot(\alpha) = \frac{\cos(\alpha)}{\sin(\alpha)}$$

$$\sec(\alpha) = \frac{1}{\cos(\alpha)}, \quad \csc(\alpha) = \frac{1}{\sin(\alpha)}$$

Complex numbers

Assume $z = x + iy \in \mathbb{C}$:

$$i = \sqrt{-1}, \quad i^2 = -1$$

$$|z|^2 = z^* z = x^2 + y^2 = \Re\{z\}^2 + \Im\{z\}^2$$

$$z = x + iy = r e^{i\theta} = r (\cos \theta + i \sin \theta)$$

$$z^* = x - iy = r e^{-i\theta} = r (\cos \theta - i \sin \theta)$$

$$|z|^2 = z^* z = x^2 + y^2 = \Re\{z\}^2 + \Im\{z\}^2$$

$$\theta = -i \ln(z/|z|) = \arctan(y/x)$$

Operations:

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

$$\cos(\theta) + i \sin(\theta) = e^{i\theta}$$

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin(\theta) = \frac{e^{i\theta} - e^{-i\theta}}{2i}$$

$$\cosh(\theta) = \frac{e^\theta + e^{-\theta}}{2}, \quad \sinh(\theta) = \frac{e^\theta - e^{-\theta}}{2}$$

Derivative: rules

Sum rule

$$(f + g)' = f' + g'$$

Scalar mult.

$$(af)' = af'$$

Product rule

$$(fg)' = f'g + fg'$$

Quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Chain rule

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Standard derivatives

Right-hand-side is the derivative of the left-hand-side

$$x^b = b x^{b-1}$$

$$\ln |x| = \frac{1}{x}$$

$$e^x = e^x$$

$$a^x = a^x \cdot \ln a$$

$$\log_b x = \frac{1}{x \cdot \ln b}$$

$$\sin x = \cos x$$

$$\cos x = -\sin x$$

$$\tan x = \sec^2 x$$

$$\sec x = \sec x \cdot \tan x$$

$$\csc x = -\csc x \cdot \cot x$$

$$\sinh x = \cosh x$$

$$\cosh x = \sinh x$$

$$\tanh x = (\operatorname{sech} x)^2$$

$$\arcsin x = \frac{1}{\sqrt{1-x^2}}$$

$$\arccos x = -\frac{1}{\sqrt{1-x^2}}$$

$$\arctan x = \frac{1}{1+x^2}$$

Anti-derivatives: rules

Fund. law $f(x) = \left(\int_b^x f(t) dt \right)'$

Partial int. $\int f dg = f g - \int g df$

$$\int f(x) g'(x) dx = f(x) g(x) - \int g(x) f'(x) dx$$

Leibniz $\frac{d}{d\nu} \left[\int f(x; \nu) dx \right] = \int \frac{\partial f(x; \nu)}{\partial \nu} dx$

Standard anti-derivatives

Rational functions:

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{1}{x} dx = \ln |x|$$

$$\int (x+b)^n dx = \frac{(x+b)^{n+1}}{n+1}$$

$$\int x(x+b)^n dx = \frac{(x+b)^{n+1}}{n+1} \frac{x(n+1)-b}{n+2}$$

$$\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \arctan(x/a)$$

$$\int \frac{x}{x^2+a^2} dx = \frac{1}{2} \ln(x^2+a^2)$$

$$\int \frac{x^2}{x^2+a^2} dx = x - a \arctan(x/a)$$

$$\int \frac{x^3}{x^2+a^2} dx = \frac{1}{2} [x^2 - a^2 \ln(x^2+a^2)]$$

$$\int \frac{1}{(x^2+1)^2} dx = \frac{1}{2} \left(\arctan(x) + \frac{x}{x^2+1} \right)$$

$$\int \frac{x}{(x^2+1)^2} dx = -\frac{1}{2} \frac{1}{x^2+1}$$

$$\int \frac{x^2}{(x^2+1)^2} dx = \frac{1}{2} \left(\arctan(x) - \frac{x}{x^2+1} \right)$$

$$\int \frac{x^3}{(x^2+1)^2} dx = \frac{1}{2} \left(\frac{1}{x^2+1} + \ln(x^2+1) \right)$$

$$\int \frac{1}{(x+a)(x+b)} dx = \frac{1}{b-a} [\ln(x+a) - \ln(x+b)]$$

$$\int \frac{x}{(x+a)^2} dx = \frac{a}{x+a} + \ln(x+a)$$

Trigonometric:

$$\int \cos(x) dx = \sin(x)$$

$$\int \sin x dx = -\cos x$$

$$\int \sin^2 x dx = \frac{x}{2} - \frac{1}{4} \sin 2x$$

$$\int \sin^3 x dx = -\frac{3}{4} \cos x + \frac{1}{12} \cos 3x$$

$$\int \cos^2 x dx = \frac{x}{2} + \frac{1}{4} \sin 2x$$

$$\int \cos^3 x dx = \frac{3}{4} \sin x + \frac{1}{12} \sin 3x$$

$$\int \sin x \cos x dx = -\frac{1}{2} \cos^2 x$$

$$\int \cos^2(x) \sin^2(x) dx = \frac{x}{8} - \frac{1}{32} \sin 4x$$

$$\int \cos^n(ax) \sin(ax) dx = -\frac{1}{a(n+1)} \cos^{n+1}(ax)$$

$$\int \cos(ax) \sin^n(ax) dx = \frac{1}{a(n+1)} \sin^{n+1}(ax)$$

$$\int x \sin(ax) dx = \frac{1}{a^2} \sin(ax) - \frac{x}{a} \cos(ax)$$

$$\int x \cos(ax) dx = \frac{1}{a^2} \cos(ax) + \frac{x}{a} \sin(ax)$$

Exponentials:

$$\int e^{bx} dx = \frac{1}{b} e^{bx}$$

$$\int x e^{bx} dx = \left(\frac{x}{b} - \frac{x}{b^2} \right) e^{bx}$$

$$\int x^2 e^{bx} dx = \left(\frac{x^2}{b} - \frac{2x}{b^2} + \frac{2}{b^3} \right) e^{bx}$$

$$\int x^3 e^x dx = (x^3 - 3x^2 + 6x - 6) e^x$$

Definite integrals

$$\int_0^1 (1-x)^m x^n dx = \frac{n! m!}{(n+m+1)!}$$

$$\int_0^1 \sin(m\pi x) \sin(n\pi x) dx = \frac{1}{2} \delta_{mn}$$

$$\int_0^1 x \sin^2(m\pi x) dx = \frac{1}{4}$$

$$\int_0^1 x \sin(m\pi x) dx = -\frac{(-1)^m}{\pi m}$$

$$\int_0^1 x^2 \sin(m\pi x) dx = \frac{(-1)^m (2 - \pi^2 m^2) - 2}{\pi^3 m^3}$$

$$\int_0^1 x^2 \sin^2(m\pi x) dx = \frac{1}{6} - \frac{1}{4\pi^2 m^2}$$

$$\int_0^1 x \sin(\pi x) \sin(3\pi x) dx = 0$$

$$\int_0^1 x^2 \sin(\pi x) \sin(3\pi x) dx = \frac{3}{16\pi^2}$$

$$\int_0^1 x^3 \sin(\pi x) \sin(3\pi x) dx = \frac{9}{32\pi^2}$$

Exponentials $x \in [0, +\infty[$

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty e^{-ax^2} dx = \frac{\sqrt{\pi}}{2\sqrt{a}}$$

$$\int_0^\infty x e^{-ax^2} dx = \frac{1}{2a}$$

$$\int_0^\infty x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{4a^{3/2}}$$

$$\int_0^\infty x^3 e^{-ax^2} dx = \frac{1}{2a^2}$$

$$\int_0^\infty x^4 e^{-ax^2} dx = \frac{3\sqrt{\pi}}{8a^{5/2}}$$

Exponentials $x \in]-\infty, +\infty[$

$$\int_{-\infty}^\infty e^{-ax^2} dx = \frac{\sqrt{\pi}}{\sqrt{a}}$$

$$\int_{-\infty}^\infty x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{2a^{3/2}}$$

$$\int_{-\infty}^\infty x^4 e^{-ax^2} dx = \frac{3\sqrt{\pi}}{4a^{5/2}}$$

$$\int_{-\infty}^\infty x^n e^{-ax^2} dx = 0 \quad \text{for } n \text{ odd}$$

Integration in spherical coordinates

$$\int_{-\infty}^\infty dx \int_{-\infty}^\infty dy \int_{-\infty}^\infty dz f(x, y, z) = \int_0^\infty d\rho \int_0^\pi d\theta \int_0^{2\pi} d\phi \rho^2 \sin \theta F(\rho, \theta, \phi)$$

where volume element $dx dy dz = \rho^2 \sin \theta d\theta d\phi d\rho$

$x = \rho \sin(\theta) \cos(\phi)$, $y = \rho \sin(\theta) \sin(\phi)$, $z = \rho \cos(\theta)$

